



Wireless Technology

Questions

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ANSWERS

Q1. Differentiate between Bluetooth and ZigBee

Feature	Bluetooth	ZigBee
Data Rate	High (up to ~3 Mbps)	Low (~250 kbps)
Power Consumption	Higher, not ideal for long-term	Very low, optimized for battery
Range	Short (10–100 m)	Moderate (10–100 m, mesh extends)
Network Topology	Point-to-point, star	Mesh, star, tree (flexible)
Applications	Audio streaming, file transfer, peripherals	Home automation, sensor networks, IoT
Cost & Complexity	Higher cost, more complex	Low cost, simple devices
Scalability	Limited number of devices	Supports large device networks

Q2. List Bluetooth Physical layer specifications

- **Frequency Band:** Operates in the 2.4 GHz ISM band.
- **Channel Bandwidth:** 79 channels, each 1 MHz wide.
- **Modulation Techniques:** Uses GFSK (Gaussian Frequency Shift Keying); newer versions also support $\pi/4$ -DQPSK and 8-DPSK.
- **Data Rate:** Up to 1 Mbps (Basic Rate), 2–3 Mbps (Enhanced Data Rate).
- **Transmission Power:** Three classes — Class 1 (100 mW, ~100 m range), Class 2 (2.5 mW, ~10 m range), Class 3 (1 mW, ~1 m range).
- **Duplex Scheme:** Time Division Duplex (TDD) with Frequency Hopping Spread Spectrum (FHSS).
- **Error Correction:** Uses Forward Error Correction (FEC) and Automatic Repeat Request (ARQ).

Q3. Differentiate between WEP and WPA

Feature	WEP (Wired Equivalent Privacy)	WPA (Wi-Fi Protected Access)
Encryption Method	RC4 stream cipher	TKIP (Temporal Key Integrity Protocol), later AES

Feature	WEP (Wired Equivalent Privacy)	WPA (Wi-Fi Protected Access)
Key Length	40/104 bits (weak)	128-bit or higher (stronger)
Security Level	Low, easily cracked	Higher, improved protection
Authentication	Shared key, static	Dynamic key management, stronger authentication
Vulnerability	Susceptible to attacks (IV reuse, weak keys)	More resistant, but WPA2/WPA3 are stronger
Usage Today	Obsolete, insecure	Still used, but WPA2/WPA3 preferred

Q4. List security requirements of Wireless networks

- **Authentication:** Ensure that only authorized users and devices can access the network.
- **Confidentiality:** Protect data from eavesdropping using encryption techniques.
- **Integrity:** Guarantee that transmitted data is not altered or tampered with during communication.
- **Availability:** Keep the network accessible and reliable, preventing denial-of-service attacks.
- **Access Control:** Restrict resources and services to legitimate users only.
- **Non-repudiation:** Ensure that a sender cannot deny sending a message, supporting accountability.

Q5. How is security provided in GSM?

In GSM, security is provided through authentication, encryption, and temporary identity protection, ensuring confidentiality and preventing unauthorized access.

- **Authentication:**

- The network verifies the subscriber using a secret key (K_i) stored in the SIM card and the Authentication Center (AuC).
- A random challenge (RAND) is sent, and the SIM computes a response (SRES) using the A3 algorithm.
- **Encryption (Ciphering):**
 - User data and signaling are encrypted over the air interface using the A5 stream cipher.
 - This prevents eavesdropping on calls and messages.
- **Key Generation:**
 - A session key (K_c) is derived from K_i and RAND using the A8 algorithm.
 - K_c is used for encryption during the communication session.
- **Temporary Mobile Subscriber Identity (TMSI):**
 - Instead of transmitting the permanent IMSI, GSM uses TMSI to protect user identity and location privacy.
- **Access Control:**
 - Only authenticated and valid subscribers can access network services, preventing unauthorized usage.

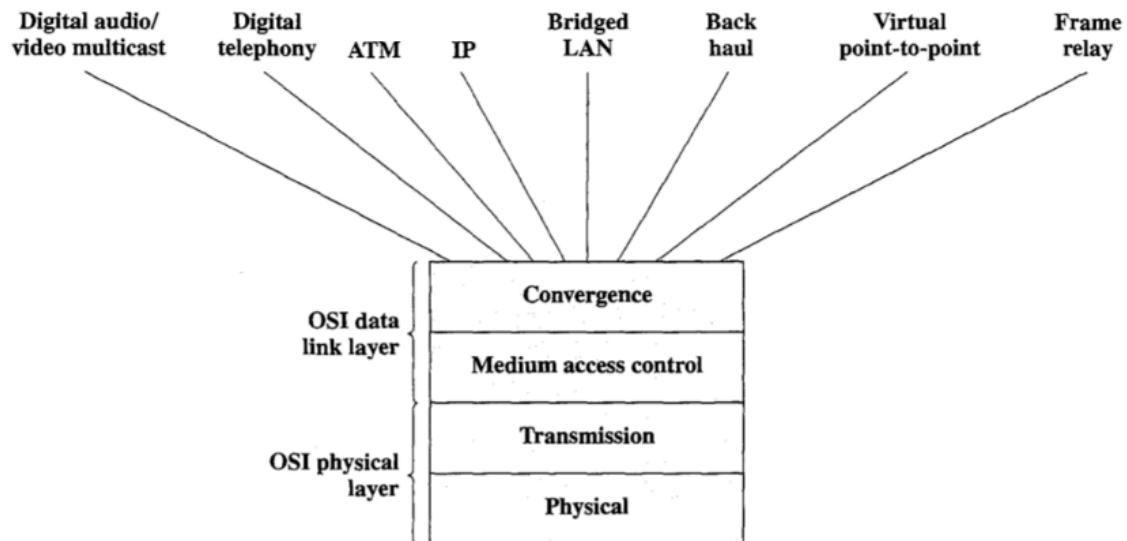
Q6. Discuss security procedure in UMTS

- **Mutual Authentication:**
 - Unlike GSM, UMTS authenticates both the user and the network.
 - Prevents fake base station attacks.
- **Authentication Vectors (AVs):**
 - Generated by the Authentication Center (AuC).
 - Contain RAND (random challenge), XRES (expected response), CK (cipher key), IK (integrity key), and AUTN (authentication token).
- **Encryption (Confidentiality):**

- Uses Cipher Key (CK) to encrypt user data and signaling.
- Protects against eavesdropping.
- **Integrity Protection:**
 - Integrity Key (IK) ensures signaling messages are not altered.
 - Provides message authentication codes (MAC).
- **Temporary Identities:**
 - UMTS uses Temporary Mobile Subscriber Identity (TMSI) and other mechanisms to protect user identity.
- **Stronger Algorithms:**
 - Uses KASUMI block cipher for encryption and integrity, stronger than GSM's A5/1.

Q7. Draw and explain protocol architecture of IEEE 802.16

Protocol Architecture of IEEE 802.16



Layers:

1. Physical Layer (PHY)

- Handles transmission over air interface.
- Supports multiple modulation schemes (QPSK, 16-QAM, 64-QAM).
- Provides error correction and channel coding.

2. MAC Layer (Medium Access Control)

- Divided into three sublayers:
 - **Convergence Sublayer (CS):** Maps higher-layer data (IP, ATM, Ethernet) into MAC service data units.
 - **Common Part Sublayer (CPS):** Core MAC functions like bandwidth allocation, connection setup, QoS.
 - **Security Sublayer:** Provides authentication, encryption, and key management.

3. Higher Layers

- Interfaces with external networks (IP, ATM).
- Ensures service delivery to applications.

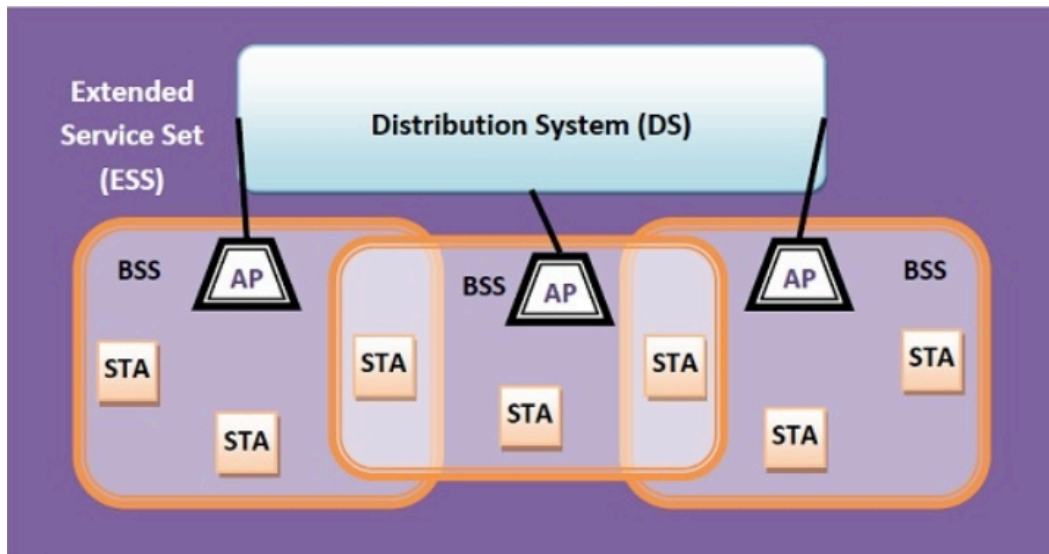
Q8. Draw and explain system architecture of IEEE 802.11. Also discuss on power management in IEEE 802.11 infrastructure network

System Architecture of IEEE 802.11

Main Components:

- **Station (STA):** Any device with a wireless NIC (laptop, phone).
- **Access Point (AP):** Connects wireless stations to the wired network.
- **Basic Service Set (BSS):** Group of STAs controlled by one AP.
- **Extended Service Set (ESS):** Multiple BSSs interconnected via a Distribution System (DS).
- **Distribution System (DS):** Backbone network linking APs to provide seamless coverage.

Diagram:



Power Management in IEEE 802.11 Infrastructure Network

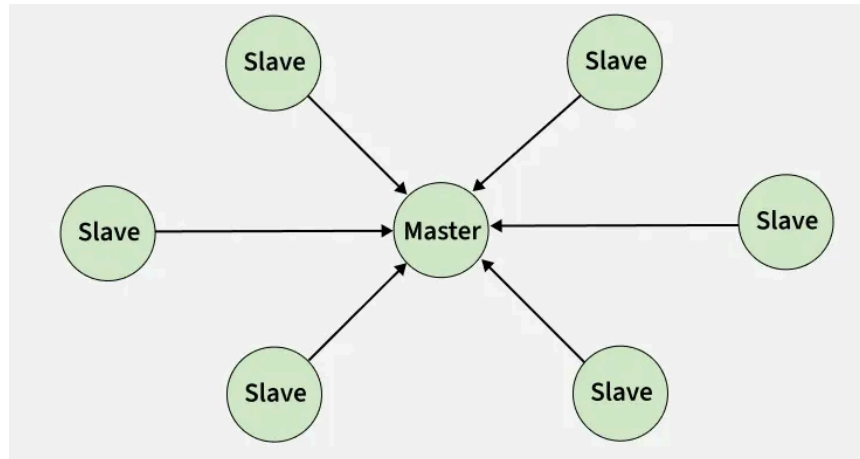
- **Sleep Mode:** Stations can enter a low-power state to save battery.
- **Beacon Frames:** AP periodically sends beacons indicating buffered data for sleeping stations.
- **Traffic Indication Map (TIM):** Part of beacon frame that tells which stations have pending data.
- **Polling Mechanism:** Sleeping STA wakes up, checks TIM, and requests data from AP.
- **Delivery Traffic Indication Message (DTIM):** Used for multicast/broadcast traffic delivery.
- **Benefit:** Extends battery life of mobile devices while ensuring reliable data delivery.

Q9. Describe Bluetooth architecture and protocol stack

Bluetooth Architecture

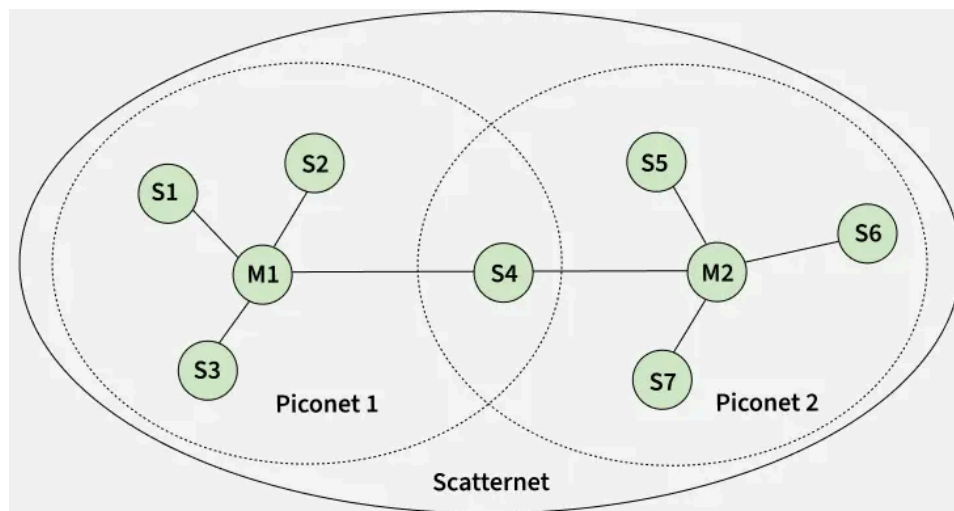
- **Piconet:**

- Basic Bluetooth network with one master and up to seven active slaves.
- Master controls timing and communication; slaves respond only when permitted.
- Uses frequency hopping to reduce interference.



- **Scatternet:**

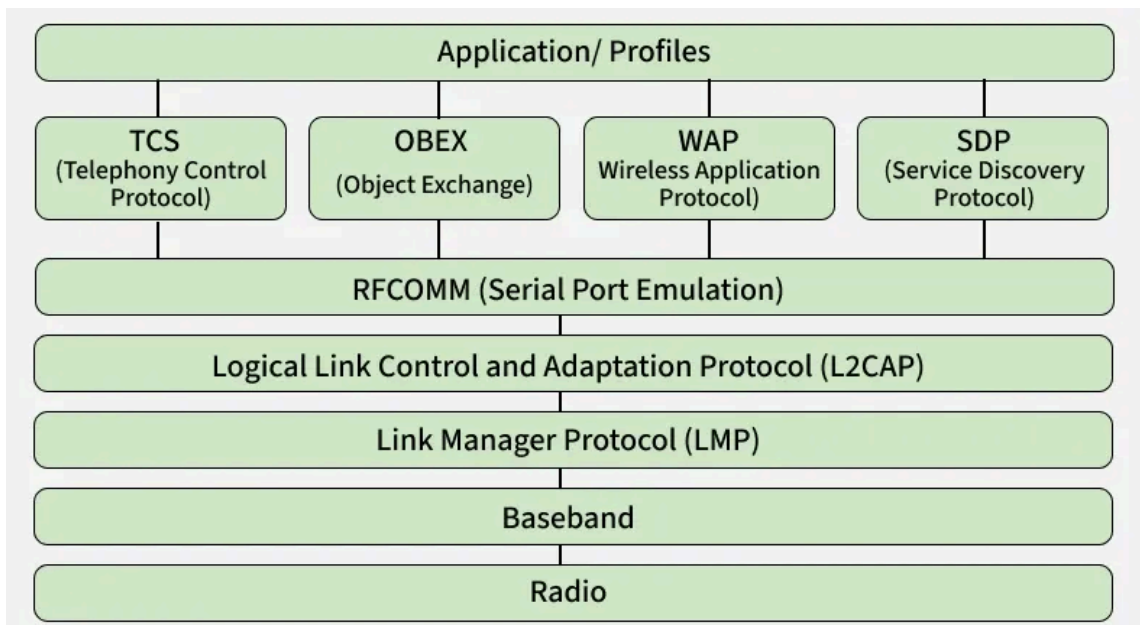
- Formed by interconnecting multiple piconets.
- A device can act as master in one piconet and slave in another.
- Extends coverage and supports communication across multiple groups.



Bluetooth Protocol Stack

- **Radio Layer:** Defines air interface, frequency hopping, modulation, and transmit power.
- **Baseband Layer:** Handles connection setup, addressing, packet formats, and timing.
- **Link Manager Protocol (LMP):** Manages links, authentication, encryption, and security.
- **Host Controller Interface (HCI):** Standard interface between hardware and host software.
- **Logical Link Control and Adaptation Protocol (L2CAP):** Provides multiplexing, segmentation, and reassembly of packets.
- **Service Discovery Protocol (SDP):** Enables devices to discover available services.
- **RFCOMM:** Emulates serial ports for data communication.
- **OBEX/WAP/TCP:** Higher-level protocols for file transfer, web access, and telephony services

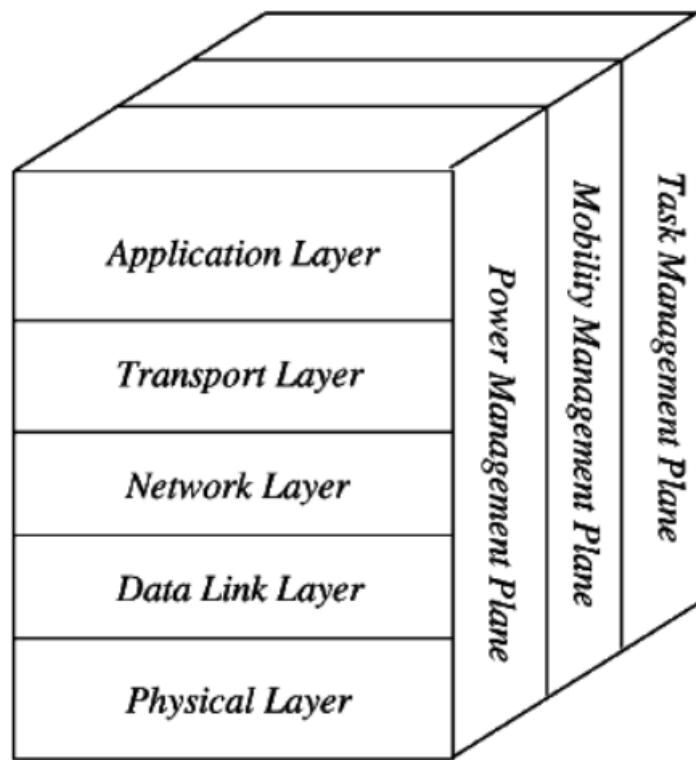
Protocol Stack (Diagram):



Bluetooth Protocol Stack

Q10. Explain layered architecture of WSN protocol and discuss issues and challenges in WSN

Layered Architecture of WSN Protocol



Wireless Sensor Network Architecture

- **Application Layer**
 - Provides application-specific services (e.g., monitoring, tracking).
 - Handles data formatting and user interface.
- **Transport Layer**
 - Ensures reliable data delivery.
 - Uses congestion control and error recovery mechanisms.
- **Network Layer**
 - Handles routing of data packets.

- Focuses on energy-efficient, multi-hop communication.
- **Data Link Layer (MAC)**
 - Provides channel access, error detection, and frame synchronization.
 - Manages collisions and ensures efficient bandwidth use.
- **Physical Layer**
 - Deals with modulation, transmission, and reception of signals.
 - Operates in ISM bands with low power radios.

Issues and Challenges in WSN

- **Energy Efficiency:** Limited battery power in sensor nodes.
- **Scalability:** Large number of nodes must be managed efficiently.
- **Security:** Protecting against eavesdropping, data tampering, and unauthorized access.
- **Data Aggregation:** Avoiding redundancy while maintaining accuracy.
- **Latency & QoS:** Meeting real-time requirements in critical applications.
- **Hardware Constraints:** Limited processing, memory, and communication capabilities.

Q11. Compare VANET and MANET

Feature	VANET (Vehicular Ad-hoc Network)	MANET (Mobile Ad-hoc Network)
Nodes	Vehicles equipped with wireless devices	Mobile devices (laptops, phones, sensors)
Mobility	Very high (fast-moving vehicles)	Moderate/low (pedestrian or portable devices)
Topology Change	Rapid and frequent	Relatively slower and less frequent
Communication Range	Larger (due to vehicle power and antennas)	Smaller, limited by device battery

Feature	VANET (Vehicular Ad-hoc Network)	MANET (Mobile Ad-hoc Network)
Applications	Traffic management, safety, infotainment	Military, disaster recovery, personal networking
Challenges	High mobility, frequent disconnections	Limited resources, scalability, energy efficiency
Infrastructure Dependency	Often integrates with roadside units (RSUs)	Purely infrastructure-less, peer-to-peer

Q12. Significance of WEP protocol. What are the features of WPA2?

Significance of WEP (Wired Equivalent Privacy)

- First security protocol for IEEE 802.11 wireless networks.
- Designed to provide confidentiality similar to wired LANs.
- Uses RC4 stream cipher for encryption.
- Provides basic authentication and access control.

Features of WPA2 (Wi-Fi Protected Access 2)

- **Strong Encryption:** Uses AES (Advanced Encryption Standard) with CCMP (Counter Mode with Cipher Block Chaining Message Authentication Code Protocol).
- **Improved Authentication:** Supports IEEE 802.1X with EAP for enterprise networks.
- **Data Integrity:** Ensures packets are not altered using robust message integrity checks.
- **Key Management:** Dynamic key generation and distribution for enhanced security.
- **High Security Standard:** Considered the most secure standard until WPA3.

Q13. What is LoRaWAN

- **Definition:** A **Low Power Wide Area (LPWA)** protocol that connects IoT devices to the internet over regional, national, or global networks.
- **Based on LoRa:** LoRa is the physical radio modulation (Chirp Spread Spectrum), while **LoRaWAN is the protocol and system architecture** that manages communication.
- **Purpose:** Enables **bi-directional, secure, and scalable communication** for IoT applications like smart cities, agriculture, and industrial monitoring.

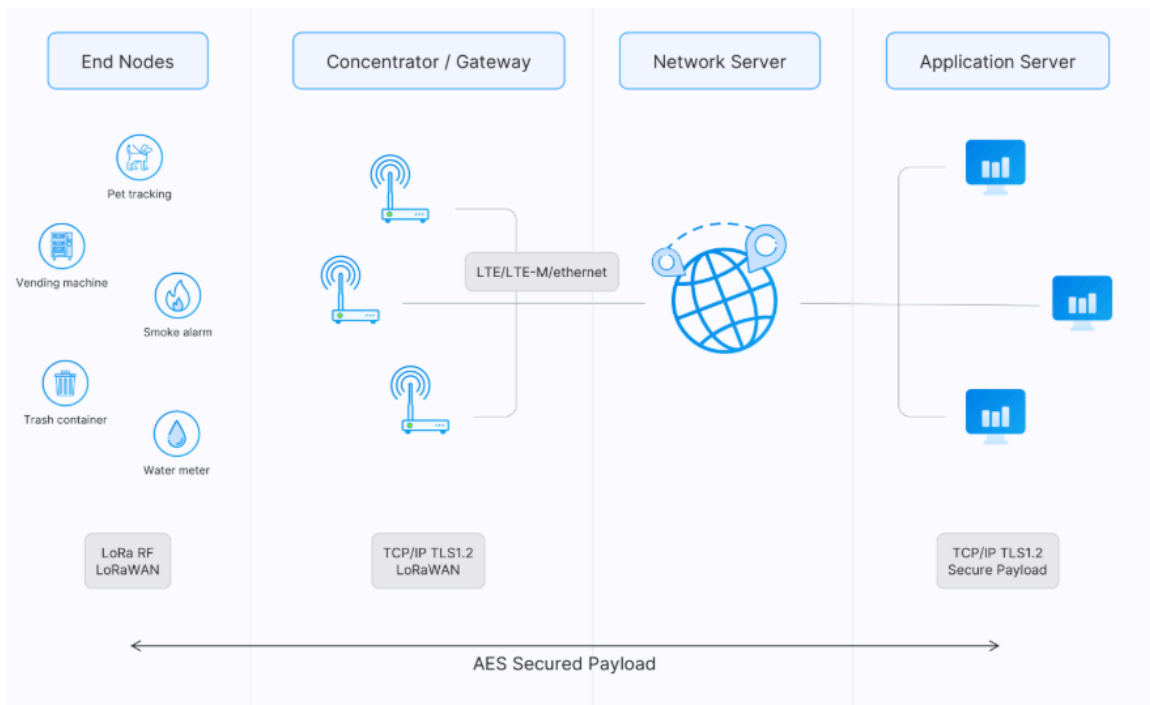
Key Features of LoRaWAN

- **Long Range:** Devices can communicate over **10–15 km in rural areas** and **2–5 km in urban areas**.
- **Low Power Consumption:** Optimized for battery-operated devices; nodes can last **years on a single battery**.
- **Scalability:** Supports networks ranging from a **single gateway to global deployments with billions of devices**.
- **Security:** Provides **end-to-end encryption** with AES keys for data confidentiality and integrity.
- **Bi-Directional Communication:** Supports both uplink (device → network) and downlink (network → device).
- **Mobility Support:** Devices can move across coverage areas while staying connected.
- **Cost Efficiency:** Low infrastructure and device cost compared to cellular IoT solutions.

LoRaWAN Architecture

- **End Devices (Nodes):** Sensors or IoT devices that send data.
- **Gateways:** Receive LoRa signals and forward them to the network server via IP.

- **Network Server:** Manages devices, security, and routing.
- **Application Server:** Processes data for end-user applications (e.g., smart meters, trackers).



Use Cases

- **Smart Cities:** Street lighting, waste management, parking sensors.
- **Agriculture:** Soil moisture monitoring, livestock tracking.
- **Industrial IoT:** Equipment monitoring, predictive maintenance.
- **Environmental Monitoring:** Air quality, flood detection

Q14. ZigBee device, topologies, architecture

ZigBee Devices

- **Coordinator:**
 - Central device that starts the network.

- Manages routing, security, and network information.
- **Router:**
 - Forwards data between devices.
 - Extends network coverage and supports multi-hop communication.
- **End Device:**
 - Simple device (sensor/actuator).
 - Communicates only with parent (router/coordinator).
 - Low power consumption, often battery-operated.

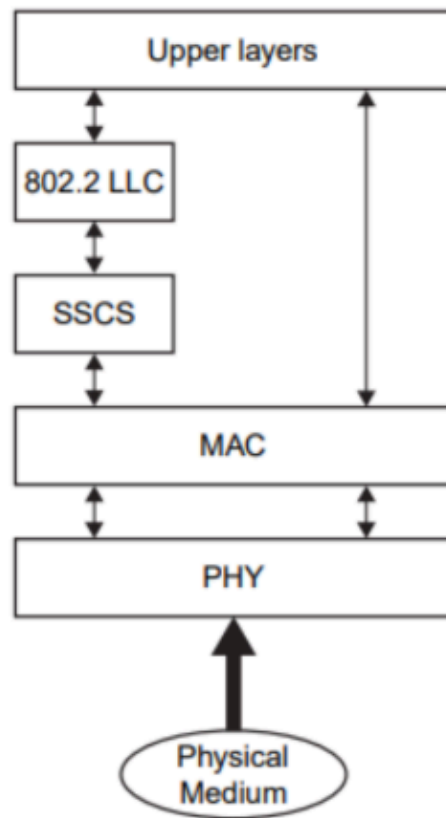
ZigBee Topologies

- **Star Topology:**
 - Coordinator at the center; end devices connect directly.
 - Simple, low cost, but limited range.
- **Tree Topology:**
 - Hierarchical structure with coordinator → routers → end devices.
 - Supports better scalability.
- **Mesh Topology:**
 - Devices interconnect with multiple paths.
 - Provides high reliability and fault tolerance.

ZigBee Architecture

- **Application Layer:**
 - Provides application profiles (e.g., home automation, health monitoring).
- **Application Support Sublayer (APS):**
 - Interfaces between application layer and network layer.
- **Network Layer:**

- Handles routing, addressing, and network formation.
- **MAC Layer:**
 - Provides channel access, frame validation, and reliable transmission.
- **Physical Layer (PHY):**
 - Operates in 2.4 GHz ISM band.
 - Handles modulation (O-QPSK) and radio transmission.



LLC: Logical link control

MAC: Media access control

SSCS: Service specific convergence
sublayer